

System Architecture, Multi-Role Login & Access Guide

Comprehensive technical reference for Patient, Doctor, Diagnostic Lab, and Hospital workflows.

PROJECT: Digital Health Platform	AUTH METHOD: Free Email SMTP OTP
TECH STACK: Django / DRF + PostgreSQL	CONSTRAINTS: Zero ABHA / Zero Docker (Native venv)

1. Core Architectural Pillars

- **Zero External ABHA / ABDM Dependency:** The platform operates 100% independently as your private digital health record cloud. There is no government API integration, no ABDM sandboxing, and no compliance delay.
- **100% Free Email SMTP Authentication:** No paid SMS gateways (Twilio, MSG91, Fast2SMS). The Django backend dispatches secure, time-limited (5-min) 6-digit OTP codes straight to the user's email inbox using Gmail App Password or custom SMTP.
- **Direct Native Execution (No Docker):** Backends runs natively with Python venv and standard local PostgreSQL/SQLite, enabling fast debugging, simple deployment, and zero virtualization overhead.
- **Frontend-First Alignment:** The database models and endpoints directly mirror the state and models of the live Flutter application.

2. Multi-Role Comparison & Differentiation Matrix

A single, clean login page accepts any registered email address. The user's role in the database governs their credentials, data accessibility, and automatically directs them to their customized portal:

Role	Primary Purpose	Signup Credentials	Landing Portal
Patient	Personal & family health locker, review lab test parameters, monitor vitals, access emergency medical card.	Full Name, Email, Gender, Blood Group (optional).	Patient Dashboard /dashboard
Doctor	Review patient directory, clinical chart history, allergy warnings, enter primary diagnosis, issue digital prescriptions (Rx).	Full Name, Email, Medical Council Reg No., Specialization, Clinic/Hospital.	Doctor Portal /doctor
Diagnostic Lab	Manage incoming diagnostic test order queue, input observed parameter values, attach report PDFs, auto-notify patients.	Lab Name, Email, Lab Registration / License No., Address, Incharge Name.	Lab Portal /lab
Hospital Care	Monitor inpatient ward bed occupancy (ICU, General, Private), admit emergency patients, finalize discharge summaries.	Hospital Name, Email, Hospital Reg ID, Wards / Departments, Staff Name.	Hospital Portal /hospital

3. Detailed Portal Breakdown & Capabilities

■ Patient Portal Capabilities

- **Health Vitals Tracker:** Records Blood Pressure (systolic/diastolic), Heart Rate, Fasting Sugar, and BMI with status tags.
- **Medical Records Locker:** Categorized repository for past prescriptions, consultation summaries, and medical history.
- **Lab Reports Table:** Diagnostic parameter viewing (e.g. HbA1c, Lipid panel) showing measured values and reference ranges.
- **Chronological Health Timeline:** Vertical timeline showing all medical interactions, visits, and hospital admissions across time.
- **Family Health Profiles:** Dedicated dependents management for children and elderly parents without requiring separate phone accounts.
- **Instant Emergency Card:** Rapid one-tap emergency card displaying Blood Group, Allergies (e.g. Penicillin), and Emergency Contacts.

■ Doctor Portal Capabilities

- **Appointment Queue:** Daily schedule showing booked patients, appointment time slots, and consultation statuses.
- **EMR Directory Search:** Live search across patient records by name, email, or system-generated Patient ID.
- **Clinical Chart View:** Full review of patient vitals trends, prior prescriptions, and diagnostic lab reports.
- **Prominent Allergy Warnings:** High-visibility warning chips alerting the clinician to severe drug allergies before prescribing.
- **Digital Prescription Generator:** Fast medicine entry with dosage (e.g. 1-0-1), frequency, and duration (e.g. 14 Days).
- **Instant Locker Sync:** Saved diagnoses and prescriptions automatically appear in the patient's mobile app timeline.

■ Diagnostic Laboratory Portal

- **Pending Test Order Queue:** Filterable by discipline (Biochemistry, Endocrinology, Cardiology, Diabetology, Nephrology).
- **Structured Parameter Entry:** Input observed numerical values, units (mg/dL, cells/mcL), and biological reference intervals.
- **Report PDF Upload:** Ability to attach certified lab report PDFs alongside structured parameters.
- **Automated Publishing:** Submitting test results immediately unlocks the report in the patient's personal locker.

■ Hospital Inpatient Care Portal

- **Ward Bed Occupancy:** Real-time bed occupancy stats across General Wards, ICU, Private, and Semi-Private rooms.
- **Inpatient Admission Intake:** Bed allocation, attending physician assignment, and primary admission diagnosis recording.
- **Discharge Summary Flow:** Formulate finalized discharge documentation with treatment summary, instructions, and follow-up date.

4. Authentication Architecture: Free Email SMTP OTP

Instead of paid SMS gateways, authentication uses a standard email OTP mechanism powered by free Gmail SMTP (or custom SMTP):

Step	Action	Backend Logic
1	User Enters Email	Frontend validates email format and sends request to <code>/api/auth/request-otp/</code> .
2	Generate 6-Digit OTP	Django generates a cryptographically random 6-digit code with a 5-minute expiration timestamp.
3	Dispatch SMTP Email	Django sends HTML email directly to the user's inbox using free SMTP settings.
4	User Enters OTP	Flutter app accepts 6 digits and submits to <code>/api/auth/verify-otp/</code> .
5	Verify & Issue JWT	If code is valid, backend generates JWT access & refresh tokens.
6	Role-Based Redirection	Existing users immediately land on their role dashboard; new users complete role registration.

5. Industry Benchmarks & References

- **Practo (practo.com)**: Patients browse records and book tests on practo.com; doctors log into **Practo Pro** (Ray) for digital clinic EMR and prescriptions.
- **Eka Care (eka.care)**: Consumer app serves as a digital health locker; **Eka Doctor** provides clinicians with digital prescription pads.
- **Apollo 24|7 (apollo247.com)**: Patient app displays Apollo records and lab tests; hospital doctors access internal HIS systems.

6. Next Implementation Steps

Step 1: Configure SMTP credentials (email address + 16-character Google App Password) in Django settings.py.

Step 2: Create Django accounts app with custom User model, roles, and OTP generation logic.

Step 3: Build /api/auth/request-otp/ and /api/auth/verify-otp/ endpoints.

Step 4: Wire the Flutter signup_screen.dart and otp_entry_screen.dart to the live endpoints.